THE PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

LONAE PRE-MOCK JOINT EXAMINATION

FORM FOUR

BASIC MATHEMATICS MARKING SCHEME

CODE; 041
TIME; 3 HRS
INSTRUCTIONS

1. This paper consists of section A and B with total of 14 questions
2. Answer all questions in both sections *A* and *B*
3. Mathematical table may be used if necessary

SECTION A: (60 Marks)

Answer all questions in this section

1. (a) $let\ X = 0.\dot{2}\dot{5}$

$100x = 25$

$100x$

$\frac{x}{99x} = \frac{0.\dot{2}\dot{5}}{25}$

$\frac{99x}{99} = \frac{25}{99}$

$x = \frac{25}{99}$ ……. (a) **01 MARK**

$let\ X = 0.\dot{5}$

$10x = 5$

$10x$

$\frac{x}{9x} = \frac{0.\dot{5}}{5}$

$\frac{9x}{9} = \frac{5}{9}$

$x = \frac{5}{9}$ ……. (b) **01 MARK**

$\frac{a}{b} = \frac{25}{99} \div \frac{5}{9}$

$= \frac{25}{99}\ x\ \frac{9}{5} = \frac{5}{5}$ **01 MARK**

(b) i 0.0021 to $2sf$ **01 MARK**

ii 8 to the nearest whole number **01 MARK**

iii 2.8 to 1dp **01 MARK**

2. (a) $\frac{2}{2\sqrt{3}+\sqrt{2}} \times \frac{2\sqrt{3}-\sqrt{2}}{2\sqrt{3}-\sqrt{2}}$ **01 MARK**

$= \frac{4\sqrt{3}-2\sqrt{2}}{(2\sqrt{3})^2-(\sqrt{2})^2}$

$= \frac{4\sqrt{3}-2\sqrt{2}}{12-2}$

$= \frac{4\sqrt{3}-2\sqrt{2}}{10} = \frac{2\sqrt{3}-\sqrt{2}}{5}$

$= \frac{2\sqrt{3}-\sqrt{2}}{5}$ **02 MARKS**

(b) $\frac{2^{3x+7}}{4^{2x-2}} = \frac{8^{x-3}}{32^{x-5}}$

$\frac{2^{3x+7}}{2^{2(2x-2)}} = \frac{2^{3(x-3)}}{2^{5(x-5)}}$ **02 MARKS**

$2^{3x+7-(4x-4)} = 2^{3x-9-(25-5x)}$

$-x + 11 = 8x - 34$

$x = 5$ **01 MARK**

3. (a)

i) $M' = \{f, b, k, c, j\}$

$N = \{f, b, k, d, h\}$ **$01\frac{1}{2}$ MARKS**

$\therefore \quad M' \cap N = \{f, b, k\}$

ii) $n(M' \cup N') =?$

But $M' = \{f, b, k, c, j\}$ and $N' = \{a, e, i, c, j\}$

$(M' \cup N') = \{f, b, k, c, j, a, e, i, c, j\}$ **$01\frac{1}{2}$ MARKS**

$\therefore n(M' \cup N') = 8$

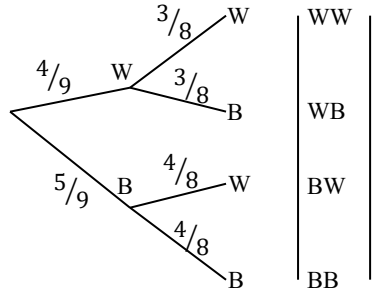
(b) let the Black Bak be B

the white Bak be W.

P(W) $= {}^{4}/_{9}$

P(B) $= {}^{5}/_{9}$

By tree diagram.



1MARK

(i) The probability that both are white balls $P(WW) = {}^{4}/_{9}\, x\, {}^{3}/_{8} = {}^{1}/_{6}$ **1MARK**

(ii) The probability that the first is black and the second is the white ball

$P(BW) = {}^{5}/_{9}\, x\, {}^{4}/_{8} = {}^{5}/_{18}$

The probability that the first is black ball and the second is the white is ${}^{5}/_{18}$

1MARK

4. (a)

i. Solve for V=$2\underline{a} + 3\underline{b}$

V=$2(3\underline{i} + 4\underline{j}) + 3(-2\underline{i} + 6\underline{j})$

V=$6\underline{i} + 8\underline{j} - 6\underline{i} + 18\underline{j}$

V=$0\underline{i} + 26\underline{j}$ =$26\underline{j}$ **02 MARKS**

ii. Find magnitude of V

|V|=$\sqrt{0^2 + 26^2}$=26 **01 MARK**

(b) (i)

Slope1 = Slope2

$\frac{5-3}{k+2} = \frac{2-0}{6-k}$

(k+2) x 2 = 2 x (6-k)

2k+4 = 12-2k

4k=8 **$01\frac{1}{2}$ MARK**

K=2

(ii)

$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

$d = \sqrt{(9-1)^2 + (5+1)^2}$ **$01\frac{1}{2}$ MARK**

$d = \sqrt{64 + 36}$

d= 10

5. (a) $2w = l + 3$……(i)

$36 = 2l + 2w$

$\frac{36}{2} = \frac{2}{2}(l + w)$

$18 = l + w$ ……(ii) **01 MARK**

Solve equation (i) and (ii)

$+\begin{cases} 2w - l = 3 \\ w + l = 18 \end{cases}$ **01 MARK**

$w = 7$ and $l = 11$

∴ width is $7cm$ and length is $11cm$ **01 MARK**

(b) $(n-2)180° = sum\ of\ interior\ angle$

$(n-2)\frac{180°}{180°} = \frac{1080°}{180°}$ **02 MARKS**

$n - 2 = 6$

$n = 8$ **01 MARK**

∴ Number of sides is 8

6. (a) $y \propto \frac{x-1}{x+d}$

$y = \frac{k(x-1)}{x+d}$

$k = \frac{y(x+d)}{x-1}$

$x = 2, d = 4, y = 1$

$k = \frac{1(2+4)}{2-1}$

$k = 6$ **01 MARK**

$\therefore y = \frac{6(x-1)}{x+d}$

From $y = \frac{k(x-1)}{x+d}$

$y = 2, d = 1, k = 6$

$2 = \frac{6(x-1)}{x+1}$ **01 MARK**

$2(x + 1) = 6(x - 1)$

$2x + 2 = 6x - 6$

$4x = 8$

$\therefore x = 2$ **01 MARK**

(b) How many grams are there in 0.00912 tones.

From 1 tonne = 1000kg **02 MARK**

0.00912 tones = 0.00912 x 1000kg

= 9.12 kg

But 1kg = 1000g

9.12kg = 9.12kg x1000

=9120 grams **01 MARK**

7. (a) $L:M = 2:1$ and $M:N = 3:4$

Required $N:L$

$\frac{L}{M} = \frac{2}{1}$ …….. (i)

$\frac{M}{L} = \frac{3}{4}$……..…. (ii)

Make M the subject in equation (i) and substitute into (ii)

$M = \frac{L}{2}$ **02 MARKS**

$\frac{L}{2N} = \frac{3}{4}$

$\frac{N}{L} = \frac{2}{3}$

$N:L = 2:3$

$\therefore N:L = 2:3$ **01 MARK**

(b) Define the following; (i) Double entry (ii) Transaction (iii) Assets **@ 01 MARK**

i. Double entry is the process of recording business transaction twice

ii. Transaction is the movement of money or goods or services valued in terms of money worth from one part/ to another.

iii. Assets are possession of a business.

8. (a) Solution

Given $A_5 = 40$ $A_7 = 20$

d =?

$A_5 = A_1 + 4d$

$A_7 = A_1 + 6d$

$A_1 + 4d = 40$ ……..i

$A_1 + 6d = 20$ ……..ii **1MARK**

Solve i and ii simultaneously

In equation i

$A_1 = 40 - 4d$……iii

Substitute eqn iii into ii

$40 - 4d + 6d = 20$

$2d = 20 - 40$

$2d = -20$

$d = \frac{-20}{2}$

$d = -10$

Then from iii $A_1 = 40 - 4(-10) = 80$

$A_1 = 80$

∴ Common difference$= -10$ **1MARK**

b) Solution

$A_{10} = A_1 + 9d$

$A_{10} = 80 + 9(-10)$

$= 80 - 90 = -10$ **1MARK**

$\therefore A_{10} = -10$

(b) Given R = 10%

(i) After two years he got 5000

To find the amount harvested

$A = P(1 + \frac{R}{100})^n$ **0.5MARK**

$5000 = P(1 + \frac{10}{100})^2$

$5000 = P(1 + 0.01)^2$

$5000 = P(1.01)^2$

$\frac{5000}{1.01^2} = P$

$\frac{5000}{1.0201} = P$ **0.5 MARK**

4901.481/= is the amount invested. **1MARK**

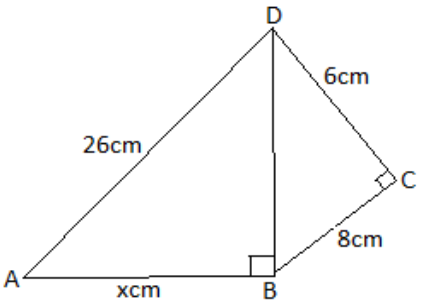
(ii) At the end of two years, he received an interest of

=5000-4901.48

=98.52/=

At the end of two years he received an interest of 98.52/= **1MARK**

9. (a) Find the length of AB in figure below



Solution

ΔBCD is a right angled triangle

$BD^2 = 8^2 + 6^2$ by Pythagoras theorem

$BD^2 = 8^2 + 6^2 = 100\ cm^2$

ΔABD is right angled

$26^2 = x^2 + 100$

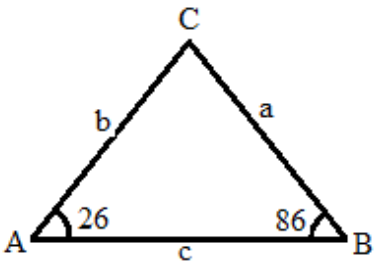
$x^2 = 676 - 100$

$x^2 = 576$

$x = 24$

$\therefore \overline{AB} = 24cm$

(b) Find unknown sides and angles in ΔABC in which $a = 22.3cm, \hat{B} = 86°$ and $\hat{A} = 26°$.



$\hat{A} + \hat{B} + \hat{C} = 180°$

$26° + 86° + \hat{C} = 180°$

$C = 68°$

From sine rule

$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

$\frac{\sin 26}{22.3} = \frac{\sin 86}{b} = \frac{\sin 68}{c}$

$\frac{\sin 26}{22.3} = \frac{\sin 86}{b}$

$b = \frac{22.3 \times 0.9976}{0.4384}$

$b = 50.7$

Again

$\frac{\sin B}{b} = \frac{\sin C}{c}$

$\frac{\sin 86}{50.7} = \frac{\sin 0.9272}{c}$

$c = \frac{50.7 \times 0.9272}{0.9976}$

$c = 47.2$

$\therefore \hat{C} = 68°, b = 50.7cm$ and $c = 47.2cm$

10. (a) $a^2 + b^2 = 60$……..(i)

$(a - b)^2 = 48$ ……(ii)

From equation ii

$(a - b)^2 = a^2 - 2ab + b^2$ **01 MARK**

$2ab = a^2 + b^2 - (a - b)^2$

Substitute the value of equation (i) and (ii) into $2ab = a^2 + b^2 - (a - b)^2$

$2ab = 60 - 48$ **01 MARK**

$ab = 6$

$\therefore ab = 6$ **01 MARK**

(b)Let the consecutive even number be x and y

The product is 80

$xy = 80$ ……(i)

The difference between the even number is 2

$y - x = 2$ ……(ii)

Then $y = 2 + x$ substitute into equation (i)

$x(2 + x) = 80$ **01 MARK**

$x^2 + 2x - 80 = 0$

Solve the value of x by splitting the middle term

$(x^2 + 10x) - (8x + 80) = 0$

$x(x + 10) - 8(x + 10) = 0$ **01 MARK**

$(x-8)(x+10)=0$

Either $x-8=0$ or $x+10=0$

$x=8$ and $x=-10$

From $y=2+x$

When $x=8 \quad y=2+8=10$

When $x=-10 \quad y=2-10=-8$

∴ Consecutive two even numbers are 8 and 10 **01 MARK**

SECTION B: (40 Marks)

Answer all questions in this section

11. (a)

02 MARKS

$C.I$	f	x	c.f	Upper r. limit
$11-15$	4	13	4	15.5
$16-20$	8	18	12	20.5
$21-25$	20	23	32	25.5
$26-30$	21	28	53	30.5
$31-35$	12	33	65	35.5
$36-40$	3	38	68	40.5
	$\sum f = 68$			

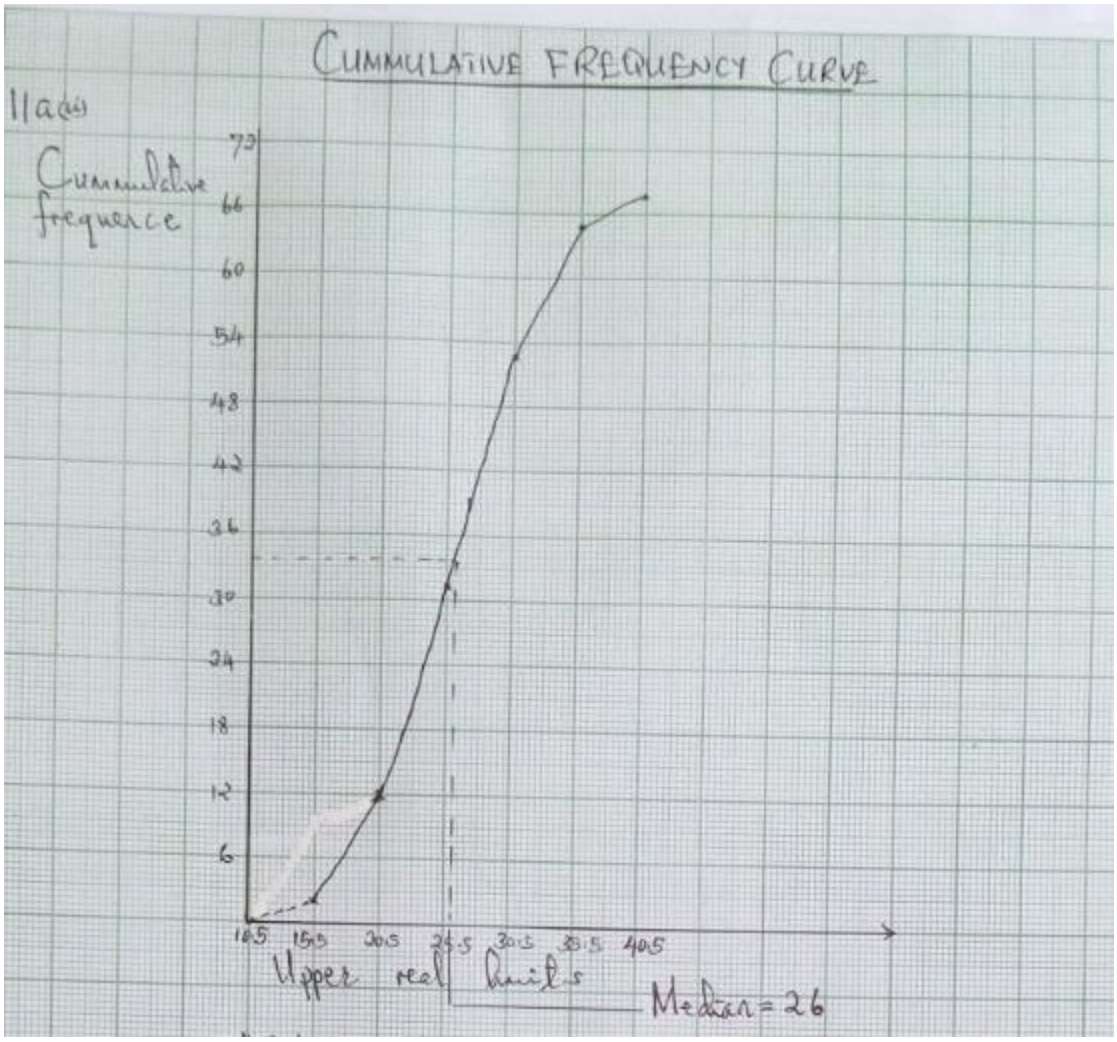
$Mode = L + \left(\frac{t_1}{t_1+t_2}\right) i$ **01 MARK**

$Mode = 25.5 + \left(\frac{1}{1+9}\right) 5$

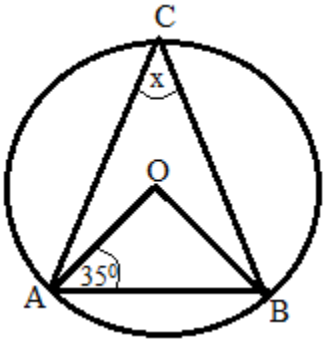
$Mode = 25.5 + \left(\frac{1}{10}\right) 5$

$Mode = 26$ **01 MARK**

(i) Diagram **03 MARKS,** Estimation **01 MARK**



(b) Find the angle x in the figure below



ΔAOB *is an Isocles*

$\overline{OA} = \overline{OB} = r\ (radii)$

$\Delta AOB = \Delta OBA$ (Base angle of an isosceles) **0.5 MARK**

$\Delta OAB = \Delta OBA = 35$

Thus $\Delta OAB + \Delta OBA + \Delta AOB = 180$

$35 + 35 + \Delta AOB = 180$

$\Delta AOB = 110$ **0.5 MARK**

But $\Delta AOB = 2\Delta ACB$

$\Delta ACB = x$

$\Delta AOB = 2x$

$\frac{110}{2} = \frac{2x}{2}$ x = 55°

Therefore the value of $\Delta x = 55^o$ **1 MARK**

12. (a)Given $B = 30^O W \quad \theta = 40^O S,\ R = 6376km\pi = 3.14\ and\ distance = 1000km$

Distance $= \frac{(\alpha - \beta)\pi R Cos\theta}{180}$ **0.5MARK**

1000km $= \frac{(\alpha - 30^o)x3.14x6370kmxCos40}{180}$ **1MARK**

$\alpha - 30^o = \frac{1000km\ x180}{3.14x6370kmxCos40}$ **1MARK**

$\alpha - 30^o = \frac{1000km\ x180}{3.14x6370kmx0.7660}$

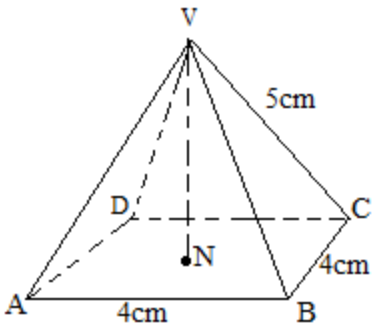
$\alpha - 30^o = 11.7^o$ **0.5 MARK**

$\alpha = 30^o + 11.7^o$

$= 41.7^o \cong 42\ W$

The location of point B is (40os 42w) **1MARK**

(b) (i) The angle between VA and the base ABCD



We need to find the distance $\overline{AN} and\ \overline{VN}$

Consider ΔABD

$\overline{AB}^2 + \overline{BC}^2 = \overline{AC}^2$

$AC = \sqrt{\overline{AB}^2 + \overline{BC}^2}$ **1MARK**

$= \sqrt{4^2 + 4^2}$

$= \sqrt{32}$

$= 4\sqrt{2}$cm **1MARK**

$AN = {}^{1}/_{2} \times AC$

$= {}^{1}/_{2} \times 4\sqrt{2}$

$AN = 2\sqrt{2}$

Consider triangle VAN

$AN^2 + VN^2 = AV^2$

$\overline{VN} = \sqrt{AV^2 - AN^2}$

$= \sqrt{5^2 - (2\sqrt{2})^2}$

$= \sqrt{25 - 8}$

$= \sqrt{17}$

$VN = 4.123$ **1MARK**

From triangle VAN, The angle VA makes with the base

$\tan\theta = \frac{VN}{AN}$

$= \frac{\sqrt{17}}{2\sqrt{2}}$

$\tan\theta = 1.4577$

$\theta = tan^{-1}(1.4577)$

$\theta = 55.6^{o}$ **1MARK**

(ii)The volume of the pyramid

$V = {}^{1}/_{3} \times Base\ area\ \times height$ **1MARK**

${}^{1}/_{3} \times 4cm\ \times 4cm \times 4.123$

$= 21.98cm^3$

The volume of the pyramid is $21.98cm^3$ **1MARK**

13. (a) Given that $A = \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} -2 & 6 \\ 1 & 2 \end{pmatrix}$.

i. 2A+B-5I=$2\begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix} + \begin{pmatrix} -2 & 6 \\ 1 & 2 \end{pmatrix} - 5\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} 4 & -6 \\ -2 & 4 \end{pmatrix} + \begin{pmatrix} -2 & 6 \\ 1 & 2 \end{pmatrix} - \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix}$

$= \begin{pmatrix} -3 & 0 \\ -1 & 1 \end{pmatrix}$ **02 MARKS**

ii. Determine the determinant of matrix in (a) (i) above

$\left|\begin{pmatrix} -3 & 0 \\ -1 & 1 \end{pmatrix}\right| = (-3x1) - (-1x0)$

$= -3$ **01 MARK**

(b) $x + 3y = 5$ and $2x + 8y = 12$

$\begin{pmatrix}1 & 3\\ 2 & 8\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix} = \begin{pmatrix}5\\ 12\end{pmatrix}$

$\begin{pmatrix}1 & 3\\ 2 & 8\end{pmatrix}^{-1} = \frac{1}{8-6}\begin{pmatrix}8 & -3\\ -2 & 1\end{pmatrix}$

$= \begin{pmatrix}4 & -3/2\\ -1 & 1/2\end{pmatrix}$ **02 MARKS**

Pre multiplying by inverse $\begin{pmatrix}4 & -3/2\\ -1 & 1/2\end{pmatrix}$

$\begin{pmatrix}4 & -3/2\\ -1 & 1/2\end{pmatrix}\begin{pmatrix}1 & 3\\ 2 & 8\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix} = \begin{pmatrix}4 & -3/2\\ -1 & 1/2\end{pmatrix}\begin{pmatrix}5\\ 12\end{pmatrix}$ **01 MARK**

$\begin{pmatrix}x\\ y\end{pmatrix} = \begin{pmatrix}4 & -3/2\\ -1 & 1/2\end{pmatrix}\begin{pmatrix}5\\ 12\end{pmatrix}$

$\begin{pmatrix}x\\ y\end{pmatrix} = \begin{pmatrix}(4x5)+(-3/2x12)\\ (-1x5)+(\frac{1}{2}x\ 12)\end{pmatrix}$

$\begin{pmatrix}x\\ y\end{pmatrix} = \begin{pmatrix}(4x5)+(-3/2x12)\\ (-1x5)+(\frac{1}{2}x\ 12)\end{pmatrix}$

$\begin{pmatrix}x\\ y\end{pmatrix} = \begin{pmatrix}20-18\\ -5+6\end{pmatrix} = \begin{pmatrix}2\\ 1\end{pmatrix}$

Therefore x=2 and y=1 **01 MARK**

(c) $\begin{pmatrix}x'\\ y'\end{pmatrix} = \begin{pmatrix}Cos\ 2x45 & Sin\ 2x45\\ Sin\ 2x45 & -Cos\ 2x45\end{pmatrix}\begin{pmatrix}4\\ 3\end{pmatrix}$ **01 MARK**

$\begin{pmatrix}x'\\ y'\end{pmatrix} = \begin{pmatrix}Cos\ 90 & Sin\ 90\\ Sin\ 90 & -Cos\ 90\end{pmatrix}\begin{pmatrix}4\\ 3\end{pmatrix}$

$\begin{pmatrix}x'\\ y'\end{pmatrix} = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix}4\\ 3\end{pmatrix}$

$\begin{pmatrix}x'\\ y'\end{pmatrix} = \begin{pmatrix}3\\ 4\end{pmatrix}$ **01 MARK**

Image after translatio

$\begin{pmatrix}x''\\ y''\end{pmatrix} = \begin{pmatrix}-2\\ 2\end{pmatrix} + \begin{pmatrix}3\\ 4\end{pmatrix}$

$\begin{pmatrix}x''\\ y''\end{pmatrix} = \begin{pmatrix}-2+3\\ 2+4\end{pmatrix}$

$\begin{pmatrix} x'' \\ y'' \end{pmatrix} = \begin{pmatrix} 1 \\ 6 \end{pmatrix}$

Hence the image is (1, 6) **01 MARK**

14. (a) A function $f(x) = \begin{cases} x \ \ if -3 \leq x < 3 \\ x^2 \ if -3 \leq x < 5 \\ 5 \ if \ x \geq 5 \end{cases}$

(i) Draw the graph of $f(x)$

(ii) State the domain and range

(iii) Find $f(10)$

(iv) (a) (i) Case I

$y = x$ for $x = -3, -2, -1, 0, 1, 2, 3$

x	-3	-2	-1	0	1	2	3
$y = x$	-3	-2	-1	0	1	2	3

Case II

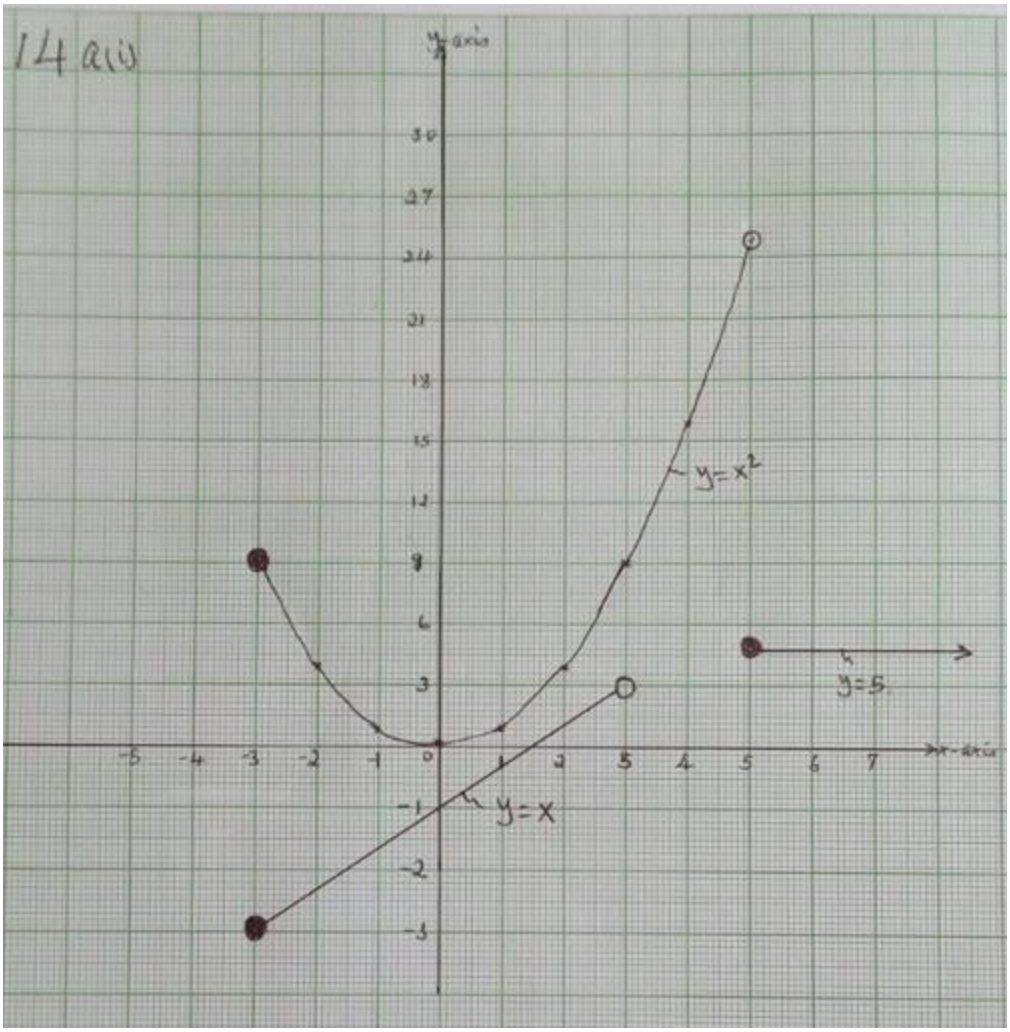
$y = x^2$ for $x = -3, -2, -1, 0, 1, 2, 3, 4, 5$

x	-3	-2	-1	0	1	2	3	4	5
$y = x^2$	9	4	1	0	1	4	9	16	25

Case II

$y = 5$ for $x = 5, 6, 7$

x	5	6	7
$y = 5$	5	6	7



02 MARK

(ii)Domain= $\{x: x \geq -3\}$ **01 MARK**

Range= $\{y: -3 \leq y < 25\}$ **01 MARK**

(iii) $f(10) = 5$ **01 MARK**

(b) Solution

Constraints

$4x + y \geq 8$

$x + y \geq 5$ **01 MARK**

$2x + 7y \geq 20$

$x \geq 0$ and $y \geq 0$

Objective function

$f(x,y) = 70x + 90y$

from (i)equation

$4x + y = 8$

Table of value

x	0	2
y	8	0

From the equation (ii)

$X + y \geq 5$

Table of value

$X + y = 5$

x	0	5
y	5	0

From the equation (iii)

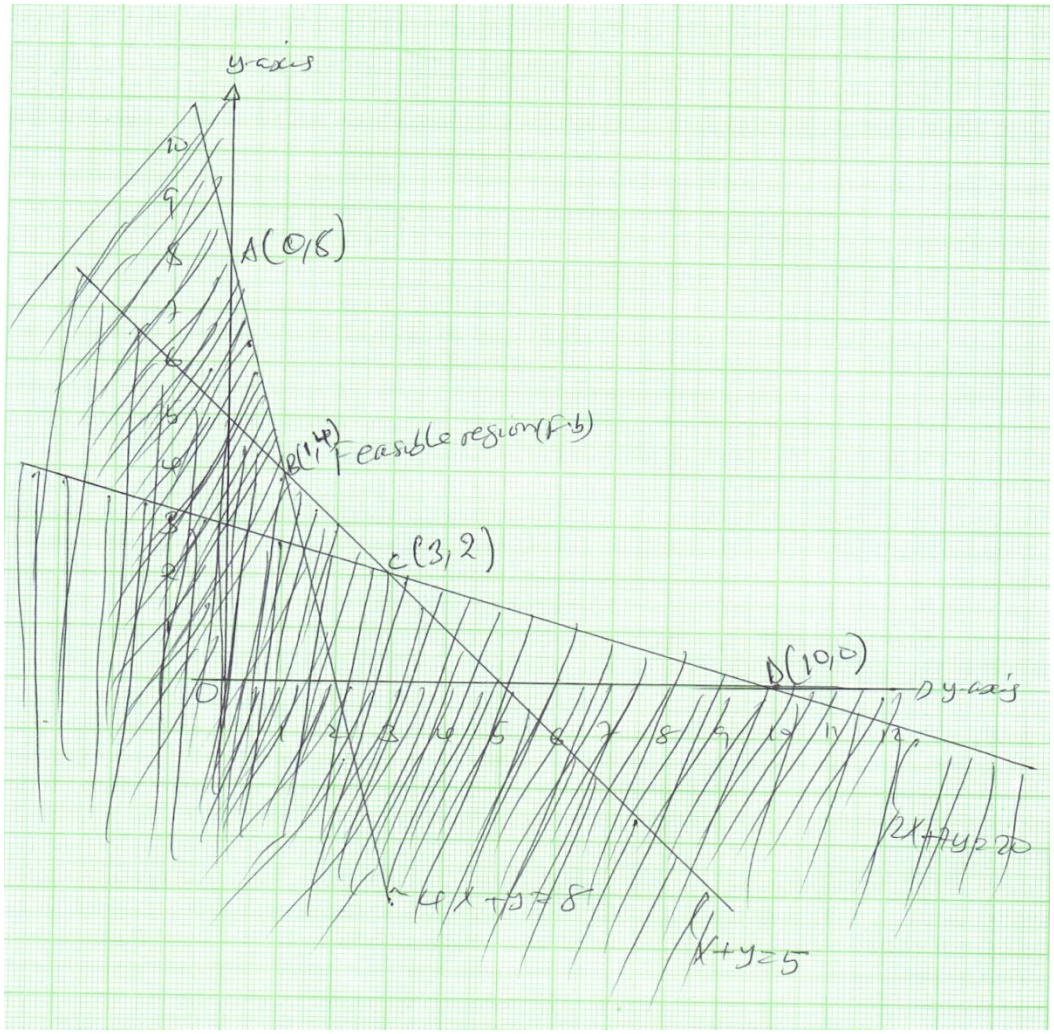
$2x + 7y \geq 20$

Table of value

$2x + 7y = 20$

x	0	10
y	2.9	0

Graph……..02 mark



refers to the graph

01 mark

Corner points	Objective functions f(x,y) =70x + 90y	Total value
A(0,8)	F(x,y)=70(0) + 90(8)	720
B(1,4)	F(x,y) =70(1) + 90(4)	350
C(3,2)	F(x,y) =70(3) + 90(2)	390
D(10,0)	F(x,y) =(70(10) + 90(0)	700

The lowest cost if the minimum of daily requirement is
350/=……..……. **01 mark**